

DHANA KARTHIKEYA VENTRAPRAGADA

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EDUCATION

University of California, Berkeley

Dec 2027

B.A. in Data Science and Applied Mathematics — GPA: 3.82/4.00

Berkeley, CA

- **Relevant Coursework:** Machine Learning · Data Engineering · Data Structures · Probability for Data Science · Principles & Techniques of Data Science · Structure and Interpretation of Computer Programs · Foundations of Data Science

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Scheme

Developer Tools: Git, GitHub, VS Code, IntelliJ, Jupyter Notebook, Google Colab, MATLAB

Libraries: pandas, NumPy, Matplotlib, seaborn, scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, SentenceTransformers

EXPERIENCE

Engineering Intern

May 2026 – Present

Guang Labs

Remote

- Building **backend AI infrastructure** for a **gaming-highlight detection platform**, supporting video ingestion, domain classification, and model-parameter selection across 6 media domains.
- Implementing a **Gemini-based media classifier** with Pydantic schemas, config parameters, and prompt-loading utilities to process images, short videos, and **10-second sampled clips** from long videos.

Undergraduate Course Staff (CS 61A) — UCS1

Jun 2026 – Present

University of California, Berkeley

Berkeley, CA

- Supporting instruction for **CS 61A** through weekly office hours, individual/group tutoring, and student support across **Python, Scheme, SQL, recursion, abstraction, and programming fundamentals**.
- Assisting course operations through **exam grading, proctoring**, student records, and focus-area support for **DSP accommodations, exam logistics, extensions, Ed/email communication**, and course infrastructure.

Data Science Intern

Jun 2026 – Present

IDX Exchange

Remote

- Building **end-to-end ML pipelines** on **large-scale MLS sold-property datasets** to predict California home prices, owning the full workflow from ingestion and cleaning through feature engineering, leakage prevention, and regression modeling.
- Iterating on **XGBoost** valuation models with cross-validation and feature-importance analysis; evaluating with **MdAPE, RMSE, MAE, and R^2** , and translating results into structured documentation for applied real estate analytics.

PROJECTS

Movie Genre Classification (Multi-Label NLP) | *Python, PyTorch, SentenceTransformers*

Jan 2026 – May 2026

- Built a **multi-label genre prediction system** on **42K+ MovieLens/TMDB films**, parsing nested IMDb metadata with regex, engineering log budget/revenue and temporal features, and embedding title+tagline+overview text via **MiniLM, MPNet, and e5-large-v2**.
- Benchmarked **Logistic Regression, XGBoost, and LightGBM** on 1024-dim e5 embeddings; combined a rare-genre cutoff with threshold tuning to address class imbalance, lifting **Macro F1 from 0.48 to 0.66** (Micro F1 0.68) with Logistic Regression beating tree models while training $\sim 12\times$ faster.

Email Spam/Ham Classifier | *Python, scikit-learn, pandas, NumPy*

Apr 2026 – May 2026

- Engineered **50+ features** from raw emails — regex counts (links, punctuation, dollar signs), HTML/subject indicators, log-scaled length features, and a curated spam/ham bag-of-words.
- Trained **L1-regularized Logistic Regression** tuned via **GridSearchCV** and validated with 10-fold cross-validation, achieving **92.1% test accuracy** on the held-out leaderboard set.

Housing Price Prediction & Assessment Bias Analysis | *Python, pandas, NumPy, scikit-learn*

Mar 2026 – Apr 2026

- Built a **linear regression pipeline** on **200K+ Cook County Assessor (CCAO) records**, engineering log-transformed sqft, **polynomial age & geo terms, latitude $\times$ longitude interactions**, and one-hot road/garage features; filtered non-arm's-length \$1 sales to align with real assessment practice.
- Achieved **test RMSE = 0.562** via 4-fold cross-validation, then quantified **systemic over- and under-estimation across price bands**, connecting design choices to documented racial inequities in property tax assessment.